

Technical Cyber security Report

**Analysis Using Splunk to create Dashboard and
Check Events logs**

By

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1. Executive Summary

- **Overview:**
 - After the proper downloading and installation of Splunk on a local computer. An analysis was conducted on event log captured on Splunk after the proper configuration of dashboard was done.
 - **Key Findings:**
 - Splunk recorded that all the activities that was been performed on the system including the clearing of events log that was done on the local computer
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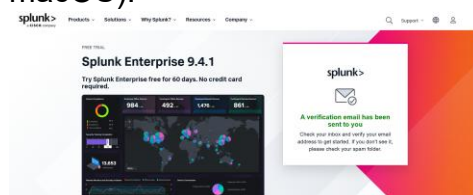
2. Background and Objectives

- **Project Context:**
 - The project aimed to check the real-time insights into the activities done on the system logs, security events, and application performance of a target system. This project was streamlined to the analysis done on the clearing of all events logs on the local computer.
 - **Objective of the Tool Use:**
 - Splunk was employed for real-time monitoring, detection, and analysis of security events to enhancement of overall cybersecurity posture through insights derived from data.
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3. Methodology

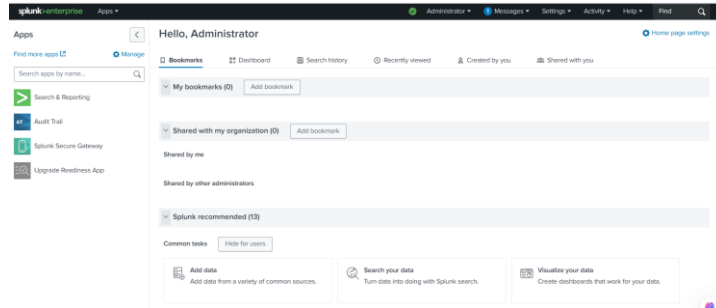
3.1 Tool Configuration

- **Downloading Splunk:**
 1. Open Kali Linux on your personal system Visit the Splunk website
 2. Click on the product – Free Trials and downloads
 3. Click on Enterprise and click on get free demo button
 4. Create an account by filling all the details
 5. After verifying the account, then download the appropriate version of Splunk for your operating system (Windows, Linux, or macOS).



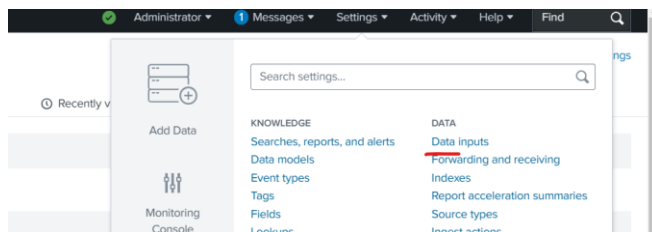
- **Installing Splunk:**

- 1) Run the downloaded installer.
- 2) Follow the installation wizard, accepting the license agreement and choosing the installation directory.
- 3) Complete the installation and launch Splunk.

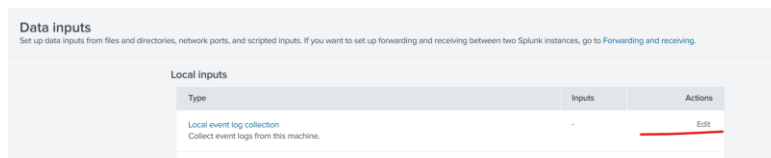


- **Configure Windows Event Log Input in Splunk:**

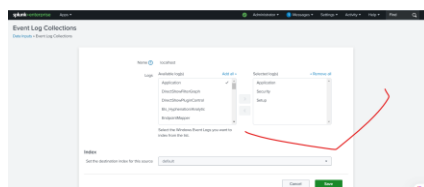
- i. In Splunk Web, go to Settings > Data Inputs.



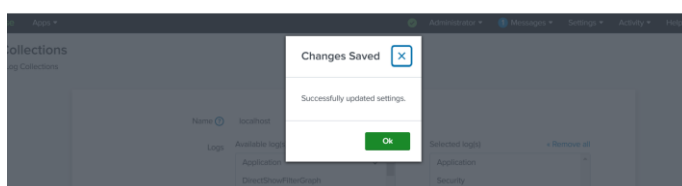
- ii. Under Local event log collection, click on Edit.



- iii. Select the event logs you want to monitor (e.g., Application, Security, System). Using the drag and drop option in arrow.



- iv. Save the configuration.

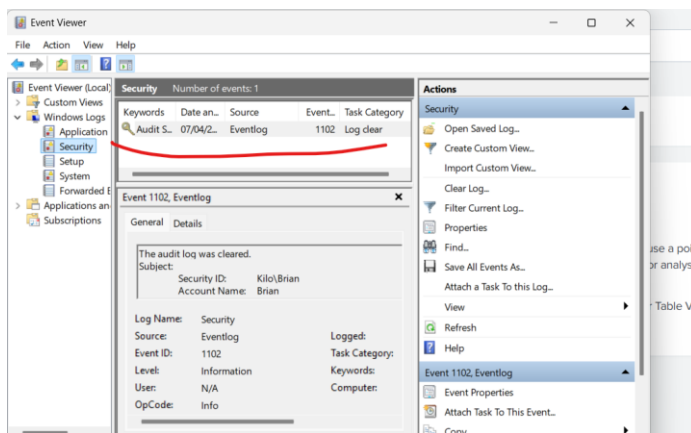


3.2 Execution Process

- **File Execution:** During the execution the following processes were carried out:

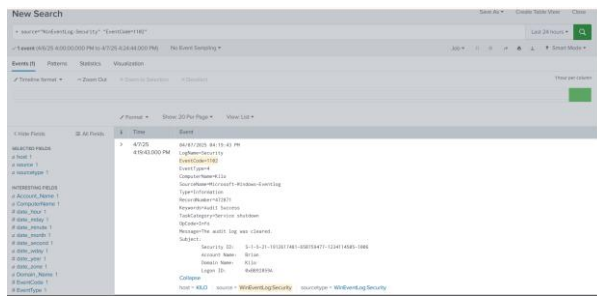
- **Clear Windows Event:**

- 1) Click on the Windows button on the keyboard
- 2) Search for event Viewer
- 3) Open Event Viewer
- 4) Navigate to the log you want to clear (e.g., Windows Logs > Security).
- 5) Right-click the log and select **Clear Log**
- 6) The Log is Cleared



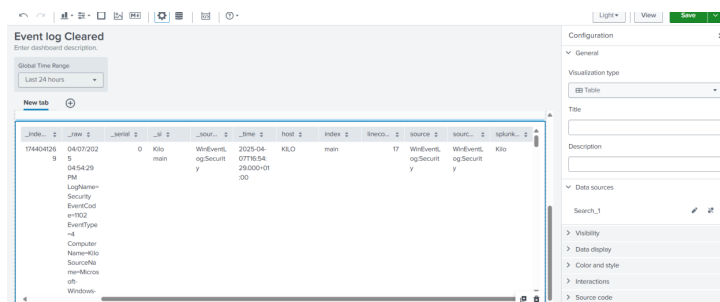
- **Applying advanced search filters in Splunk to refine event analysis:**

- 1) Click on search and Reporting
- 2) Type “*” in the search field and click search to show all the events
- 3) Left click twice on the particular thing you want to add to the filter
- 4) Click on Add to search to add it to the search filter
- 5) Do this for as many filters you want to add to it
- 6) Search for the different event code
- 7) Using the <https://www.ultimatewindowssecurity.com/> to search for the different event code
- 8) Select Event code = 1102, repeat the previous step 3-4 to add to the search filter



○ Create Dashboard on Splunk:

1. Click on Create Table View
2. Unclick the Raw option
3. Click on done button
4. Click on Dashboards
5. Click on create a new dashboard
6. Click on Table
7. Paste the search query `* source="WinEventLog:Security" "EventCode=1102"` to the SPL Query and
8. Click on save



Monitoring and Analysis:

● System Monitoring:

- The Splunk dashboard monitored in real time all the events that occurred on the target system and created a log for all the events on the system. It was able to capture the details of the clearing event using the event code - 1102

● Network Monitoring:

Splunk monitored the network activities of the local computer and recorded it as events with different code

4. Findings and Analysis

4.1 Indicators of Compromise (IOCs):

There was no IOC detected.

4.2 Risk and Impact Assessment:

● Potential Impact:

Consistent Utilization of Splunk to monitor event logs helps in the faster detection of possible security issues through real-time log analysis. Splunk provides avenue of proper root cause identification based on all the event details captured in the logs.

5. Recommendations

5.1 Immediate Remediation Actions:

- Continuous monitoring of the event clearing activities on the dashboard to ensure that the IT defensive team is immediately notified if any malicious user clears the event logs of a particular system.

5.2 Long-Term Mitigation:

- **Monitoring and Detection Enhancements:**
Recommend that there is continuous monitoring of the dashboard to ensure compliance and immediate detection of such activity in the future.
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6. Conclusion

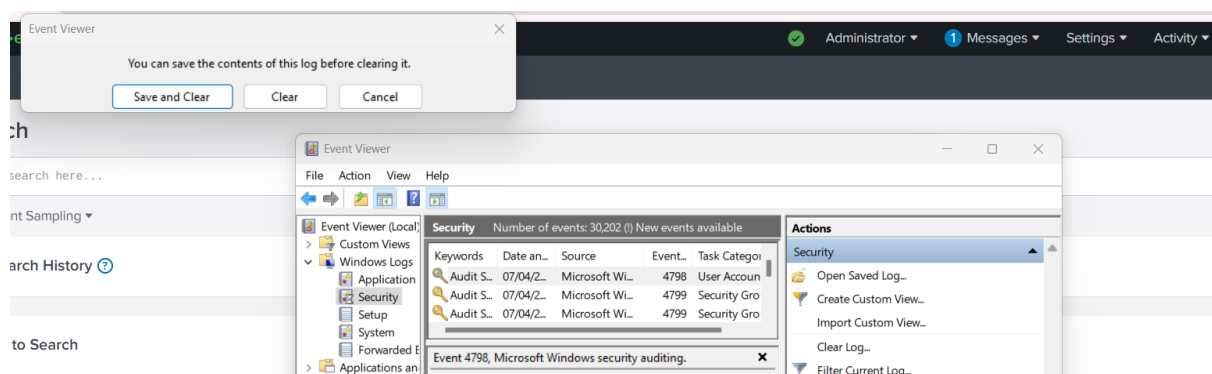
- **Summary of Findings:**

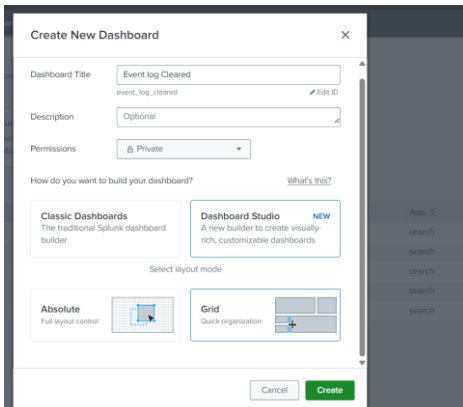
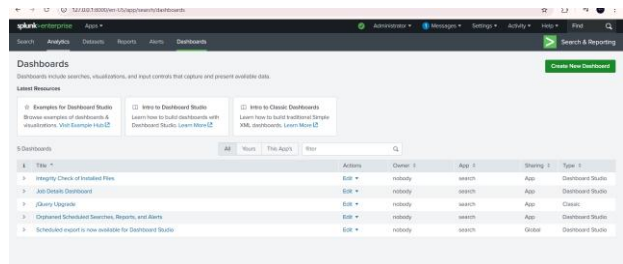
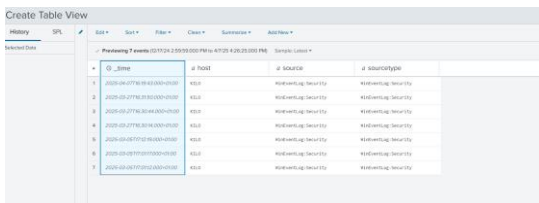
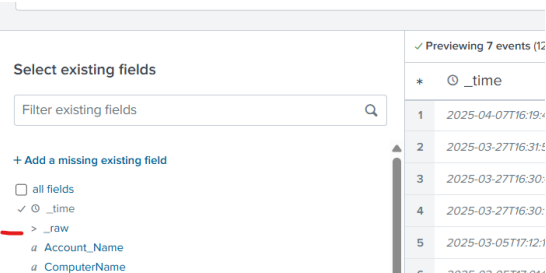
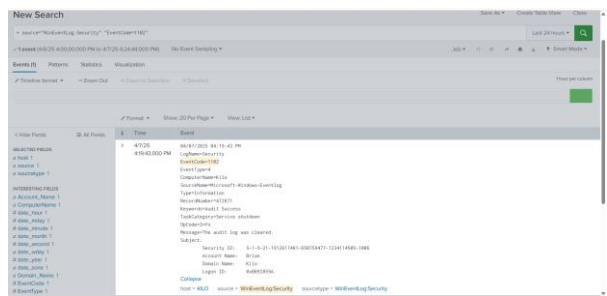
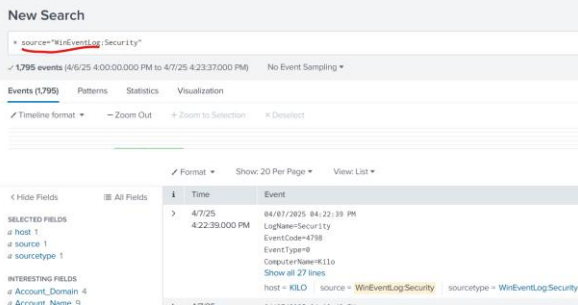
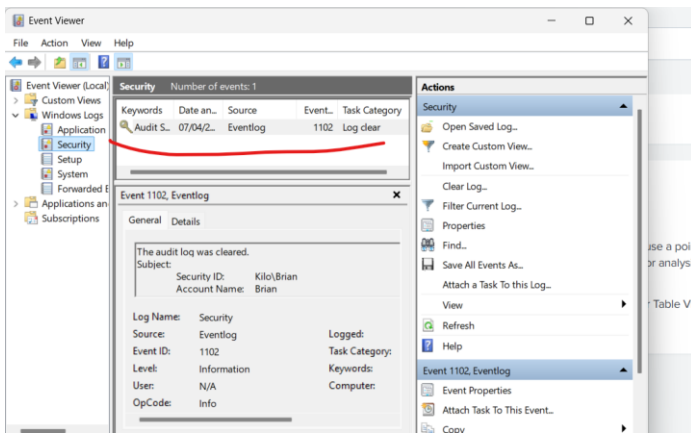
Splunk was able to provide real-time detection, streamlined response based on different events, and improved compliance through centralized log analysis and dashboard monitoring.

Appendix: Additional Data

- **Raw Logs and Tool Outputs:**

Here is the raw result from Splunk





Table

Events

Single value

Single value radial

Area

Column

Line

Bar

Pie

Bubble

Parallel coordinates

Punchcard

Scatter

Filter gauge

Marker gauge

Choropleth SVG

Map

Link graph

Search

Assets

Reports

Alerts

Light

View

Save

New data source

Data source name

Search_1

Access search results or metadata

SPL query

Open in search

source="WinEventLog:Security" "EventCode=1102"

Time range

Input

Static

Default

Time range set by dashboard source default value

\$global_time.earliest\$ - \$global_time.latest\$

Configure in source editor

1 visualization will be updated

Cancel

Apply and close

Event log Cleared

Enter dashboard description.

Global Time Range

Last 24 hours

New tab

_index	_raw	_serial	_si	_sour...	_time	host	index	lineco...	source	sourc...	splunk...
174404126	04/07/2022 04:54:29 PM	5	0	Kilo main	2025-04-07T16:54:29.000+01:00	KILO	main	17	WinEventLog:Security	WinEventLog:Security	Kilo
LogName=Security EventCode=1102 EventType=4 Computer Name=Kilo SourceName=Microsoft-Windows-											

Configuration

General

Visualization type

Table

Title

Description

Data sources

Search_1

Visibility

Data display

Color and style

Interactions

Source code